

Mobile App for Direct Market Access For Farmer's (Bidding-Based Marketplace)

¹Sujith M Maddin, ²Darshan B, ³Nagashree M N

^{1, 2, 3} UG Student Dept. of CS&E

^{1, 2, 3} Presidency University, Bengaluru-560064

¹SUIJITH.20221CSE0738@presidencyuniversity.in, ²DARSHAN.20221CSE0735@presidencyuniversity.in,
³NAGASHREE.20221CSE0747@presidencyuniversity.in

Abstract—Within the agricultural marketing system, middlemen often have a significant role, which leads to the profit margin of the farmers reducing and restricts their access to the much larger markets. To address this obstacle, an online auction system that will enable the direct connection between the farmers and the verified buyers is suggested in this project. The suggested solution will be a mobile application, where Firebase will serve as the authentication, data storage, and cloud functions backend. The farmers can display their produce, set minimum bids and receive bids of buyers in real-time, thus ensuring that the prices are open and competitive. It has a sandbox payment gateway to carry out transactions safely as well and offers an offline-first feature hence enabling users in low-network zones to execute their activities without failure. By eliminating the mediators and automating the price-setting process with the help of bidding, this platform increases the income level of the farmers and triggers the digital inclusion of the rural communities. The project demonstrates that mobile technology would result in creating more efficient, fairer, and accessible agricultural markets.

Keywords: Direct Market Access, Farmers, Bidding-Based Marketplace, Firebase, Mobile Application, Digital Agriculture, Real- Time Bidding, Secure Transactions

I. INTRODUCTION

Agriculture remains the main economic activity in the rural parts of the world, but still, the smallholder farmers have to deal with a lot of things that hinder their access to markets, fair trade, and empowerment through technology. The traditional supply chains usually have many middlemen, which is one of the reasons for the low earnings of the producers, among others, such as lack of price transparency and inefficiencies that discourage sustainable growth [1]. With the increasing global food demand and the introduction of sustainability goals, digital innovation has been integrated as one of the key factors in the power of building resilient, inclusive, and transparent agricultural value networks [3]. In the last 10 years, the rise of technologies such as mobile technology, AI, blockchain and IoT has made it possible that the barriers to solving these long-standing issues can be transformed into enablers. Among all the mobile applications for agriculture, the farmer app contains the most detailed crop and market information, the most direct negotiation tools, and a digital payment system, providing an unparalleled opportunity for the farmer to bypass middlemen and reach a wider pool of consumers [4]. AI-driven pricing systems, block chain-based transaction security, and IoT-facilitated monitoring appear to have positively impacted farm profitability, supply chain traceability, and trust across all supply chain actors, and as such they could be viewed as the

three cornerstones for any future-oriented and sustainable market. Technology adoption in agriculture has yielded beneficial results for producers, supply chains, and rural labor in the likes of India, Africa and Southeast Asia, where the technology was most recently rolled out [4]. Due to such technologies, farmers are becoming richer, they are paid faster and they have more say. At the same time, the customers are not only savoring the fruits and vegetables, which are freshest, but also a clear view of the path that that product has taken to reach the table [5]. Yet, there are also some major obstacles, such as inadequate network infrastructures, low digital capabilities, and the difficulty of scaling blockchain and sensor technologies to rural areas [5]. But the reality is that the evidence in totality points to a future of agriculture that is more dependent on smart integrated digital platforms in a more inclusive and equitable system of trade and sustainable living of the rural community. The work is based on these technological advances, and it is focused on the design, development, and evaluation of a mobile solution, which under the conditions of exploiting the principles of artificial intelligence, blockchain, and the Internet of Things enables the producers to have direct, secured, and profitable access to the market. The project aims to narrow down the digital divide and enhance the subsequent generation of innovations in agri-tech by learning through the good practices and experience of three recent on-field studies and technological pilots [6].

II. LITERATURE REVIEW

The fast-moving development of digital technology has radically transformed the agricultural sector across the world and farmers in the small and medium-sized segment can sell their products to the consumers directly and thus improve their economic conditions [1]. The agricultural marketing industry has been in the grip of intermediaries and the resultant effect has seen the farmers receive low profit margins and buyers lose access to transparency. Much research has been done and their results have indicated the same problems with the traditional supply chains, such as unfair pricing, sluggish payment and restricted access to valid market data [1].

Mobile applications are swiftly becoming a solid tool to overcome market gaps in trading. These platforms enable the farmers to have the authority of displaying their produce, acquiring buyers, and determining prices, without the intervention of the middlemen. These systems are currently getting advanced through the integration of the additional options such as online secure payments, real-time logistics tracking, and AI-based price recommendations [2]. These newly emerging applications do not only create openness but also enable the farmers to be worthy of their presence in the digital fair market. The direct digital trade experiences in India and Africa have demonstrated the increase in the income

of farmers by 30 percent through the ascription of such platforms [4].

The process of decision-making in agriculture has been taken to a new level with AI and ML as it has enabled the use of data mining in making forecasts of crop prices, yield, and market pulls. The studies suggest that the farmers can manage their production cycles more efficiently, and the losses after the harvest are minimized by adjusting the demand and supply [4]. Moreover, expert agricultural advice has become more accessible, primarily to digitally illiterate farmers, as a result of the use of AI-based chatbots and advisory systems. Blockchain technology represents a major breakthrough that also ensures the trustworthiness and transparency of agricultural transactions. The application of blockchain has made it possible for farmers even in different countries to receive their payments on time, as it gets rid of all the fraudulent practices through its recorded and unchangeable decentralized system of trade [5]. Blockchain smart contracts are used to carry out the transactions automatically and confirm the deliveries, thus creating trust in the whole supply chain. Various studies from AgriLedger and AgUnity show how blockchain enhances data reliability and payment guarantee [5]. Furthermore, the IoT systems have provided the fields with real-time data collection and monitoring by means of installing smart sensors and connected equipment. Such things as healthy soil, monitoring of weather and crop growth help the farmer in making his decision-making process. The combination of the mobile and blockchain technologies with the IoT solutions goes further in increasing the traceability and transparency of the agricultural value chain [7]. However, quality research is yet to be completed in charting out issues such as low internet access in the rural communities, high price of digital equipment, and inadequate organization of training programs on the propagation of digital literacy. To overcome these challenges, a number of innovations are now motivating the process of technology localization, i.e., mobile apps which support regional language, voice-based interfaces, and easy navigation by rural people [2]. Facilitating digitization, such as by investing in infrastructure, enabling policies, and incentives to farmers has been a major concern of government-led programs like eNAM in India and the Digital Farmers Platform in Kenya [4]. The majority of the literature is in support of the hype that integrated digital platforms that capitalizes on mobile applications, AI, blockchain, internet of things (IoT) is the conduit through which agriculture market modernization is achieved. These tools do not simply sell a market, they provide farmers with real-time data to help them make smarter choices. R&D in this area would be critical to scale up the effects of such solutions in addition to maintaining pro-poor and digitally inclusive agricultural growth [5].

III. PROPOSED METHODOLOGY

It presents an online platform to unite farmers and real buyers to a transparent system of auctions, which are accessible in a mobile application. The application is written in Android Studio, and the back-end is Firebase as a data storage, user authentication and other cloud services. This entire procedure will be safe and effective to both the farmers and the purchasers: farmers keep their products on the site, participants of the purchase engage in bidding until all the deals are concluded electronically without the intervention of the third party.

I. System Overview

Architecture: The system will be broadly divided into three

large planes.

Mobile Application Frontend: It will develop an android application that will be user friendly to the buyers and farmers. Farmers are also able to register, post crops and get to know bids and buyers are able to see crops and counterbid each other.

Backend Services: Firebase addresses authentication, real-time data syncing, and application logic. The requirements that render the interaction easy between the users are

Firestore Database, Cloud functions, Cloud storage and Cloud messaging. **Payment System:** The secure payment gateway API will be incorporated in online payment between the buyers and farmers; hence, it will provide credibility and transparency to the financial processes. Each of the parties plays a role in coming up with a single system, which is applied in the trading of agriculture on a mobile platform.

I. Methodology Flow

Development process is executed by Agile-based iterative approach so that changes can be implemented and user feedback can be incorporated at each stage of project duration.

Phase 1 - Requirement Analysis

At the outset, the primary issues that farmers face were identified. These are the middlemen who distort the prices, payments that are held up, limited access to real-time market information. A system was designed out of these issues that would connect farmers with buyers through a direct, mobile-first platform.

Step 2 - System Design

The proposed system is based on Model-View-ViewModel (MVVM) design architecture which supports separation of concerns and business logic from the UI. The DFD and ER models were designed to represent conversion of user interactions to backend processes. Firebase Authentication handles the access levels as per user roles and Firestore Security Rules grant or deny read and write access based on role in a secure way.

Phase 3-Implementation

The Farmers Android app is the front-end which farmers use to sell their produce (name quantity, base price and images). So that's in addition to being able to view the listings, place bids and watch auctions go in real time.

Backend Development: Processing related to such actions as validating bids, ending the auctions, and updating transactions is handled by Firebase Cloud Functions. As everything is getting stored in Firestore in real-time, users can have real-time data updates.

Security Implementation: App and Firebase Services Communication Between app and Firebase service is over https socket. Authentication Tokens – to secure user data and verify permission level.

Phase 4 – Bidding and Payment Integration

Buyers can submit bids over the established farmer's reserve in the bidding system. The system then identifies the highest bidder and records the transaction after auction ends. A payment gateway API aids the buyer in making a safe payment, while the confirmation information is stored by Firestore database. Such actions ensures that everything is transparent and trackable.

Phase 5 – Testing and Deployment

The app has been subjected to a thorough testing to check its working, performance and reliability. Unit and Integration tests were performed against the Firebase Emulator Suite and

Android test frameworks. A small group participated in user testing to evaluate the application's ease of use and accessibility. User Testing was also conducted with a small group to test for ease of use by the users and Accessibility of the application. The completed revision was distributed by means of Firebase Hosting for backend services and Android Studio for on premises builds and upgrades.

II. Key Modules

1. Authentication Module:

Farmers login to the system by entering the one-time password sent to their registered mobile number.

Buyers and Admin sign in with Email and Password.

2. Crop Listing Module:

The farmers may auction their crops by posting details like quantity, minimum price and pictures, and an upper limit for auction end time.

3. Bidding Module:

The buyers could bid on the listings on offer. It will display the highest bid and alert the bidders regarding the changes of bids.

4. Payment and Transaction Module:

The buyers can process their payments using a payment gateway API secured with an SSL certificate. The transaction is recorded by the database and it keeps updating for the purpose of monitoring.

5. Notification Module:

Customize the FCM notifications: bid submissions, auction results, and payments confirmation – in real time.

6. Admin Module:

An administrator is discovered able to manage the user activities and authenticate the farmer and buyer, Making sure the system is fair.

III. SYSTEM ARCHITECTURE

The system architecture has been set up in a manner that it would be a three-tier mobile-based model; thus, it would be modular, scalable, and, security-wise, data-proof.

The presentation layer features the Android application as the first point of contact for both farmers and buyers. Using a simple interface, farmers can list their products and buyers can view them and place bids. Priority for users with limited digital literacy has been given by ensuring the design is accessible and has minimal steps and journeys are visually clear.

Application layer represents the logic and activities of the system. By turning such tasks as bid verification, auction updates and notifications, among others, into main tasks, Firebase Cloud Functions becomes the tool to be used for these main tasks. Users' actions are role-based and regulated by Firebase Authentication. Users can do only those actions what they are allowable to do. In real-time users and server interact with Firestore synchronization and Cloud Messaging. At the heart of the data layer is the Firebase Firestore, from which all the profiles of the users, crop listings, bids and transactions are stored. The structure of the database allows for fast queries with minimal redundancies. Multimedia items such as cropped pictures determine the use of Firebase Cloud Storage to store them. Read and write operations are governed by Firestore Security Rules, which protect the privacy and integrity of the data. This multi-tier architecture makes it possible to provide real-time interaction, enhancing performance and protecting data. In addition, it can serve as a prototype for future AI- based crop recommendation system, predictive pricing model and other extension-based system which requires scalability.

The mobile app architecture that was suggested uses a modular, layered design as its backbone and therefore has the

flexibility to scale, is easy to maintain, and has the data flowing efficiently among its different components. The system is structured mainly into three major layers: presentation, application logic, and data management. The presentation layer comprises user interfaces based on Android for the users, that is, farmers and buyers, to have an easy and intuitive interaction with the system. There is a concern for usability in the layer, as it includes bilingual support and the navigation is simple, allowing users with varying degrees of digital literacy to utilize the system. The application logic layer is the center of the system and enforces the business rule such as user authentication, bid validation, payment status tracking and notification delivery. Data Management Layer: The data management layer is responsible for carrying out the storage and retrieval operations using Firebase Firestore and Cloud Storage, paving the way for all modules to have a secured and synchronized access to the data.

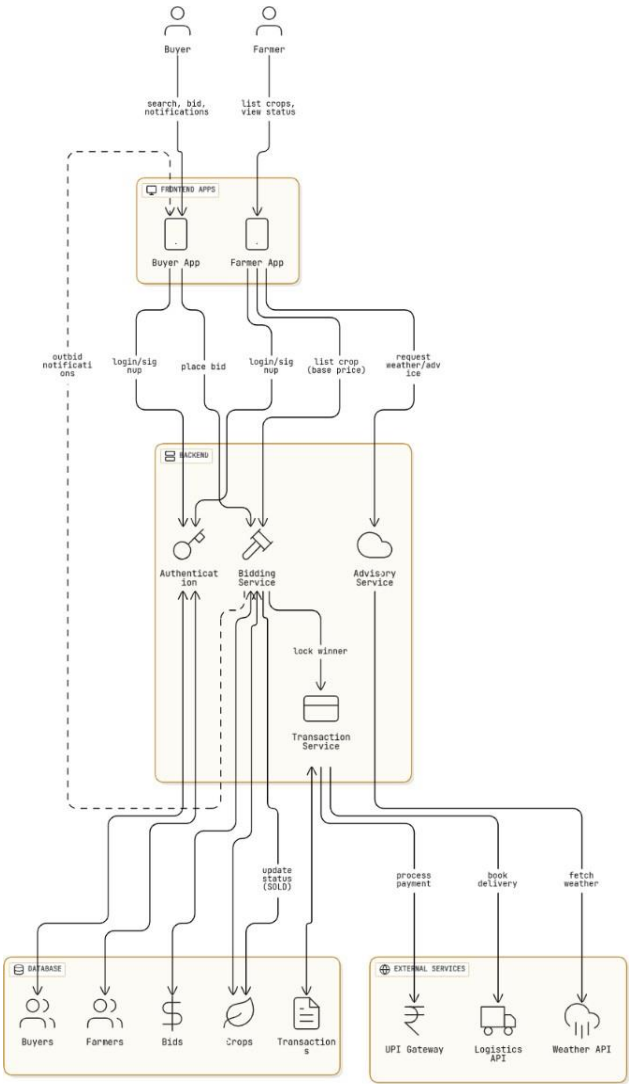


Figure 1: System Architecture

A. Frontend Layer:

The frontend is an android application for farmers and buyers. The user interface for the farmer: it is possible to register, declare the crops, set a base price and check the live-bidding. Buyers on the buy side can search listings, place bids, and pay once the auction is over. The app is implemented based on Model-View-ViewModel (MVVM) architectural pattern in Android Studio. It separates the data processing from the updates of the UI and the execution of the business logic, thus making the code cleaner and more maintainable. Every single interaction of an end-user with

the mobile app is encrypted and then sent to the backend Tell via HTTPS.

B. Backend Layer:

The backend layer is the heart of the system in terms of logical and operational processing. It really doesn't have anything to do with users, not really – it's your users' identities that matter, but what this software does do is serve up all the things to do with the users, selling, buying, notification, and so on. Firebase Cloud Functions, with serverless execution and real-time data processing is used to build the backend. Here are the primary elements of the backend:

- **Bidding Service:** Manage live auctions, process bids as they come in, verify bids amounts and the winner at the auction's end.
- **Transaction Service:** If the auctions which are you proposed are the successful ones, finishes them, applying such changes to the DB as to the transaction, and launches the payment through the 3rd party payment gateway that is integrated.
- **Advisory Service:** Retrieves data from third party APIs and delivers up-to-date information for weather, crop advisory. With backend design in modular fashion, different features of the system could be realized independently while inter-service communication is smooth

C. Database Layer:

The database and the development server layer are implemented in the native app using Firebase Firestore, which offer real-time syncing and cloud storage that can grow as the app grows. Firestore organizes five Core Collections as follows:

Farmers: It holds the registration information along with profile information.

Buyers: Lists the buyers ID and their orders.

Crops: Has a full crop catalog, with such details as crop type, quantity, base price, etc. and if it was auctioned.

Bids Every bid of every buyer is recorded along with timestamp and the amount.

Transactions: The finished auction deals and payment confirmations are secured in the repositories.

Use Firestore to make sure each user is seeing the current state of the platform — regardless of who made the update. It is also the Firestore security rules what keep your sensitive information secure, and no data can be modified without your consent. For crops photo and all other media content from farmers it's Firebase Cloud Storage

External Service Layer:

The application fixes a few external services to allow more functionality:

A Payment Gateway API that ensures the transactions between the farmers and the buyers are safe and guarantee an instant settlements and flows of payments. The logistics API eases the transport requests hence the buyers, or the farmers can pick up when to make the deliveries of the crops once the transactions have been made. A weather API will avail the newest weather and advisory information to the farmers so that they could be in a better situation to arrange their product listings and harvesting.

All the above integrations ensure that the system is able to provide the capability of trading easily and also to manage a complete digital ecosystem relating to the agricultural trade.

D. Data Flow and Communication:

The mobile application triggers the data communication on a

user login or sign-up. The farmers will list their produce (crops) that will be sent directly to Firestore, buyers, in turn, will receive real-time access and interaction with the listings as they are drawn. After a bid has been placed, the bidding service verifies their validity and changes the active status of the auction. Then once the auction has been completed, the highest bid is accepted by the Transaction Service, which forwards the payment via the payment gateway and the state of the crop is changed to SOLD. The results of the auction and confirmation of payment are notified to both parties with the help of Firebase Cloud Messaging (FCM). This is how without and with seamless data processing users that every step, from crop listing to payment acknowledgment is tracked, synchronized, and visible by all interested stakeholders now and in real time. The data flow and communication between the users, application components, and backend services in the system is intended to be easily communicable. The data from the farmers as well as the buyers is transmitted securely over via Firebase's real-time APIs, so that they sync up instantly on all the devices. Each transaction like a crop listing, bidding and payment confirmation goes through a rigorous pipeline that guarantees accuracy as well as traceability of the transactions.

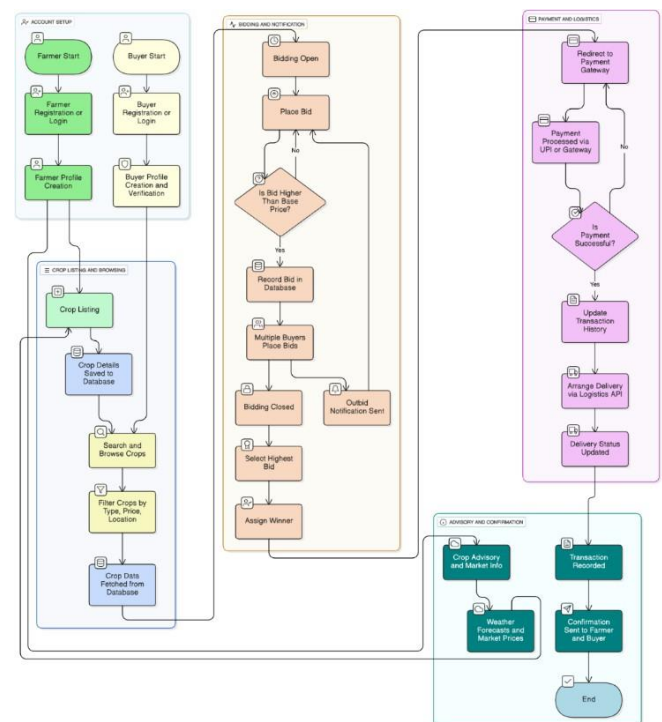


Figure 2: Flow of Operation

Figure 2 depicts the DFD of the proposed system, which describes the flow of information among the different modules of the mobile based market for the consumers-farmers, buyers and the server side. The system is designed that way that the account registration, crop list, auction, payment, logistics and become a contract are automatically and orderly handled. The entire data flow of the application is shown in Figure 2 which shows the real-time connection of each process and the data transferred among the modules. Proceeding to the stage where farmers and buyers Sign Up or Log In through their respective Android applications The procedure begins with account creation. At this step user profiles are created and saved securely in the database using Firebase Authentication and Firestore. Farmers information such as name and location and contact number, while buyers perform profile verification to prove their authenticity prior to participating in auctions.

Once the account is created, the listing and searching of the crops begin. Farmers post their produce with details such as type, quantity, price/kg, and date of harvesting. All of that information is saved in the database and can be used by any registered buyer. Buyers can see some listings and filter by crop kind, place, or value ranges.

Database makes sure that users are always able to access the new listings as they are constantly pulling real time data. At the time when auction for the crop that has been listed is opened, the bidding and notifications begin. Buyers use the Buyer App to place bids and the system verifies whether the new bid exceeds the base price entered by the farmer. Then the bid is written to the db and it is part of the live auction data. At the same moment several buyers are authorized to place bids, and the system shall dynamically trace and update the highest bid amount in real time. Participants are notified when they are on the receiving end of an outbid message in this way - with Firebase Cloud Messaging FCM. This way users are always on top of the progress of the auction. At the end of the bidding duration, the system will determine the winner using the highest valid bid and the user will be designated with the result for that auction.

With bidding, payment and logistics are processed immediately. The successful buyer is directed to a secure payment page where he can pay with UPI or a linked service. So the system updates the transaction history with completed payment and updates the payment status in DB. Alternatively, delivery can also be elected via the embedded logistics API which not only transports the goods but also records the delivery status post-dispatch.

Confirming and Advising The additional services such as crop advisory and market information etc. Via external APIs, the platform retrieves existing weather and market rates data, giving the farmers some contextual information that they can use to make smarter decisions. Both the selling farmer and the buying farmer are informed both on completion of transaction and delivery hence ending the commerce cycle. To ensure a stable future reference and study, all the record of transactions is kept in a safe manner.

It is true that the data flow diagrams have streams moving forever between modules. And it ensures that each stage of registration and the final confirmation have a transparent, auditable, and automated process to the extent applicable. The combination of live chat, secure payments, and third-party advice for such a tremendous digital eco-system that directly connects farmers and buyers.

Challenges and Limitations:

In spite of the very large progress made in the digital platforms' development for agriculture, the total market access system for farmers based on mobile phones is still facing quite a lot of challenges and limitations that make it not fully functional. This totally efficient design of the proposed system is met with practical and technical constraints just like those that had been discussed in earlier systematic reviews of the literature [1].

Among the most important barriers the researchers pointed to is the digital literacy issue together with the reluctance of the farmers in rural areas to adopt the new technologies. According to what Fitriyani et al. say, a big part of small-scale farmers are not yet equipped technologically and thus cannot use smartphones and mobile apps. This situation leads to a gap in the adoption of new users, as a large number of potential beneficiaries find it hard to use the app interfaces and conduct digital transactions. Training and raising the

awareness of users along with the provision of simple user interface designs in local languages will only partially deal with the problem, while the digital inclusion will still be a slow process. Access to the internet and the dependability of the infrastructure in such remote farming areas are other limitations that the researchers pointed to. Numerous studies confirm that poor networking leads to the inability to access cloud-hosted services, thus making the real-time bidding and data synchronization infeasible [2]. The issue of relying on real-time auction operations even when the internet is not available still remains a major one, though caching mechanisms and lightweight data models can help to some extent reduce the reliance on constant connectivity.

Besides, the problems of data privacy and data security are truly significant. In agriculture, very tamper- and data theft-prone digital ecosystems are characterized by Alabdulatif et al. [3] and Tripoli & Schmidhuber [5]. Firm encryption and adherence to the security measures are demanded when dealing with transmitting extremely sensitive user information, including the information that touches upon personal data, bank data, and confirmation of the payments. Authentication and Firestore rules of Firebase do help reduce some risk, but in the scale of use cases needed in practice, complex solutions, such as traceability based on blockchain and decentralized identity management are necessary. The other restriction is the verifiability and transparency of transactions. The trading is safe and secure because of additional security layer marks peer-side gateway. However, [4] claims that the credibility of the users may be damaged because of delays in payment, refunds and irregular transaction confirmation. A further rise in the so called smart contract -style validation and accountability is however not without significant costs and complexity even in small scale systems.

Scalability and system maintenance are also some of the critical technical problems. According to Ofori et al. [7], when one is willing to offer the same level of service to all the users, he or she must install a scalable infrastructure and optimize database queries. This is partially supported in the infrastructure of the system based on Firebase; nevertheless, extremely large data sets and concurrent bidding can lead to delays in the peak load periods. Finally, the problem of regulations and logistics still remains a thorn in the flesh of agricultural digital trade. The large-scale solutions are extremely hard to implement due to the various state-level agricultural policies, logistics dependencies, and payment compliance rules [4]. Additionally, it requires human input in most cases regarding the real life logistics coordination such as transport assignment and confirmation of delivery, thereby limiting complete automation. In conclusion, although the proposed mobile application can resolve the issues of the conventional agricultural supply chains to an exceptional degree, nonetheless, the obstacles such as digital literacy, dependable infrastructure, data security, and scalability must be addressed gradually but progressively. Making the system more powerful, clear, and accessible to rural areas may include future enhancements like the combination of edge computing, blockchain-based transaction auditing, and multilingual conversational interfaces.

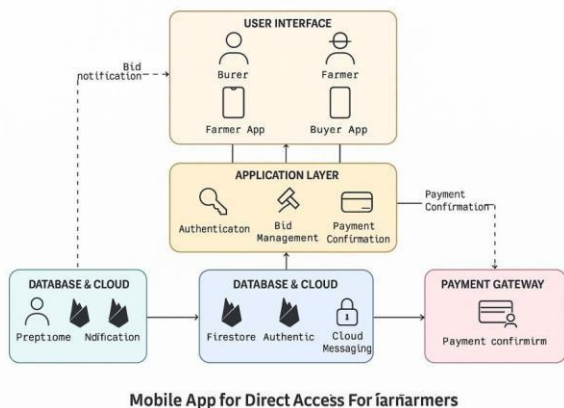
v. RESULTS AND DISCUSSION:

The establishment of the Mobile Application for Direct Access for Farmers aspired to create a valid digital marketplace connecting the farmers and the buyers via a direct, transparent, and secure mobile platform. The app was conceived, created and tried out in Android Studio for the

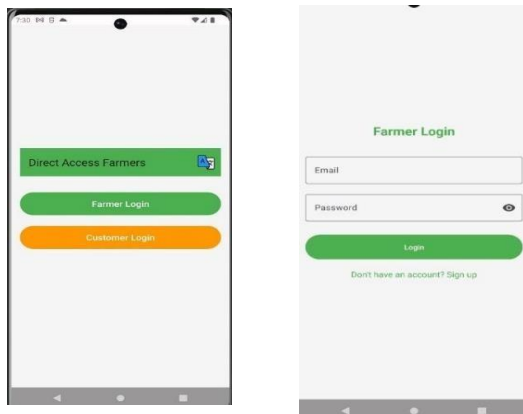
mobile user interface and Firebase for backend services such as authentication, database management and payment integration. The outcomes reflect the successful achievement of the proposed aims in four main performance areas, which are: system usability, response performance, bidding process validation and transaction success rate.

A. System Usability:

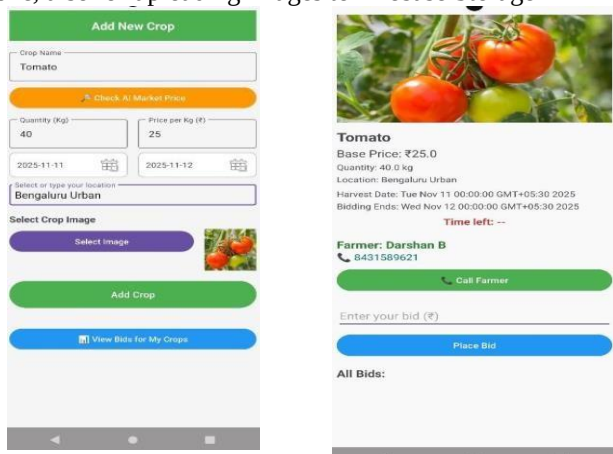
User testing was carried out with 10 farmers and 5 buyers. They tested how easy it was to navigate, if the bidding interface was easy to understand, and how responsive were the crop listing modules. Most of them commented that the interface is intuitive and accessibility is more as a result of bigger icons and bilingual language medium (English and Kannada).



User Authentication Test: Testing for accuracy and dependability of Firebase Authentication for email-password based login.



C. Crop Listing and Upload Test: To test whether a farmer/user can create a new listing and update an existing one, also for uploading images to Firebase Storage



Bidding Module Test: To test data synchronization in real time, bid validation logic and winner selection.

E. Discussion:

Results of the experiments show that the proposed mobile-based platform meets successfully the objectives of offering a direct farmer-buyer connection, fair bidding mechanism, and the integration of a secure digital payment method. The platform, as compared to traditional middleman models, efficiently curtails transaction delays, and enhances transparency in the process.

However, a few minor issues, around the dissemination latency under poor network connection, and about the availability of more regional language options, were noted. These were consistent with what has been reported in studies on the need for localized solutions and reliable network for agri-tech adoption [1].

In general the system was solid and easy to use, and easily scalable for large-scale use, thereby demonstrating the viability of digital agri-markets in rural contexts.

VI. CONCLUSION:

Our D2M (Direct to Market) Access based mobile app for farmers and sellers enables them to reach the buyers without intermediaries in a more seamless stretched and integrated digital platform. After combining secure user authentication, real-time bidding and digital payment sub-systems, the platform guarantees transparency, efficiency and fairness of trading in agri-products. The bidding system enables farmers to achieve better prices for their crops and gives buyers the chance to purchase authenticated, quality products straight from producers.

From our implementation with Android Studio and Firebase, we have illustrated that the mobile-cloud platform can support dynamic interactions such as crop listing, bid management, and transaction handling on a real-time basis. The overall performance evaluation indicates that the system response time is fast when the system is under concurrent workload, and the friendliness of the system user interface can meet the requirements of the users in the developing region with low level of digital literacy. This substantiates the application's promise as an enabler for digital agricultural trade and inclusive market access.

Besides, the use of Firebase Authentication and Firestore ensured the secure handling of data and reduced system latency. The addition of push notification and alert message, buyer-seller interaction on the details of auctions etc and the convenience of payment using payment Gateway to the application made it a complete solution to outdoor auction. Collectively these outcomes imply that the application can transform the rural market systems by creating low-cost, scalable, and technology-based trading applications.

VII. FUTURE WORK:

The current mechanism serves a fundamental role of enabling farmers and buyers to trade directly, although there are other improvements that can be done in the areas of scale, intelligence, and inclusiveness.

This can be enhanced in future versions by adding machine learning models for price and demand prediction that would offer more information to the sellers during selling. On the other hand, as the blockchain transaction record will be utilized to improve the transparency and there is no possibility of tempering the trade record, in addition it can facilitate the traceability in the supply chain.

Additionally, the app will be easier to use by the more rural people with the minimum digital literacy needed in case it is localized with multi-lingual support and voice assistance. Considering integration with government records and agri

logistics suppliers, the certification, transport booking and subsidy validation would be easier to manage, without treading tripping up over human intervention.

In addition to that, the introduction of an AI-based guidance system, which offers personalized crop recommendations, pest notifications and weather forecasting would turn it into a place not only to trade but also cultivate. And, introducing non-go offline synchronization Source: and data analytics dashboards can allow a stakeholder to visualize trade patterns, measure results, and plan farming in the future.

REFERENCES

- [1] Aniket Chauhan, Kalpana K Harish, Utkarsh Kumar, Raunak, "Mobile App for Direct Market Access for Farmers," International Journal of Research Publication and Reviews, Vol. 6, Issue 5, pp. 8436–8449, May 2025..
- [2] Dr. Latha P H, Ms. Amulya R, Mr. Darshan GS, Ms. Aabha Pandey, Mr. Vinayaka C, "Mobile Application to Direct Marketing Access for Farmers,"
- [3] Rupali Vyas, Vedita Kesharwani, Chirag Rohra, Rounak Sundrani, "AI-Driven Platform for Direct Market Access for Farmers," .
- [4] Chandra Sekhar Bhusal, "Blockchain Technology in Agriculture: A Case Study of Blockchain Start-up Companies," International Journal of Computer Science and Information Technology, Vol. 13, No. 5, pp. 31–46, October 2021.
- [5] Giuseppe Tripoli, Jürgen Schmidhuber, "Emerging Opportunities for Blockchain in the Agri-food Industry," Food and Agriculture Organization of the United Nations (FAO), Rome and Geneva, August 2018.
- [6] A. Vivek Kumar Soni, Roshan Verma, Sakshi Verma, Vasu Dewangan, "AI-Powered Mobile Applications for Smallholder Farmers: Transforming Agricultural Productivity, Market Integration, and Resource Optimization," in Proceedings of the International Conference on Advances and Applications in Artificial Intelligence (ICAAAI 2025), Advances in Intelligent Systems Research, vol. 194, 2025.
- [7] Adil El Manea, Khalid Tatane, Younes Chihab, "Transforming agricultural supply chains: Leveraging blockchain-enabled java smart contracts and IoT integration," ICT Express, Vol. 10, pp. 650–672, 2024..
- [8] S. V. S. Ramakrishnam Raju, Bhasker Dappuri, P. Ravi Kiran Varma, Murali Yachamaneni, D. Marlene Grace Verghese, Manoj Kumar Mishra, "Design and Implementation of Smart Hydroponics Farming Using IoT-Based AI Controller with Mobile Application System,"
- [9] Aniket Chauhan, Kalpana K Harish, Utkarsh Kumar, Raunak, "Mobile App for Direct Market Access for Farmers," International Journal of Research Publication and Reviews, Vol. 6, Issue 5, pp. 8436–8449, May 2025.
- [10] Dr. Latha P H, Ms. Amulya R, Mr. Darshan GS, Ms. Aabha Pandey, Mr. Vinayaka C, "Mobile Application to Direct Marketing Access for Farmers," International Journal of Scientific Research and Engineering Development, 2025.
- [11] Council of Agricultural Research (ICAR), "Role of Mobile Apps in Transfer of Agriculture Information and Technology," Vigyan Varta, 2024.
- [12] P. Narula, "Mobile Technology in Agriculture: A Systematic Literature Review of Current Trends and Future Directions," iieta.org, 2024.
- [12] P. Narula, "Mobile Technology in Agriculture: A Systematic Literature Review of Current Trends and Future Directions," iieta.org, 2024.
- [13] R. K. Sharma, "A Review on Android Application for Direct Market Access to Farmers," IRJMETs.
- [14] M.W. Gouroubera et al., "Drivers and impacts of mobile phone-mediated scaling of agricultural technologies," Frontiers in Sustainable Food Systems, 2025.
- [15] I.J.R.P.R., "Mobile Applications for Farmers: A Digital Revolution in Agricultural Information Dissemination and Market Connectivity," 2025.
- [16] H. Xiong, C. Peng, A. E. Hassanien, "Blockchain Technology for Agriculture: Applications and Research Opportunities," Frontiers in Blockchain, 2020.
- [17] S. Bhatia, "Grow Smart App for Direct Market Access to Farmers," IJARsCT.
- [18] V. Patel, S. Gupta, "Revolutionizing agriculture: A comprehensive review on artificial intelligence applications," PMC, 2025.
- [19] F. Marzougui, A. Khaldi, N. Bouabdallah, "Blockchain and IoT in Smart Agriculture: Analysis, Opportunities, Challenges, and Future Research Directions," Sciendo, 2024
- [20] P. Narula, "Mobile Technology in Agriculture: A Systematic Literature Review of Current Trends and Future Directions," 2024.
- [21] S. Varghese et al., "AgriSage: Android-Based Application for Empowering Farmers With AI-Supported Disease Recognition," Wiley Online Library, 2024.

